Kamran Amini

Senior Systems & Solutions Architect
Architecture & Technology Advisor to CTOs

Amsterdam, The Netherlands - Vienna, Austria
Nationality: Dutch
+31 6 11060215
Email: kam.cpp@gmail.com
kam@dekrypt.nl
Website: dekrypt.nl
GitHub: github.com/kamcpp
LinkedIn: linkedin.com/in/kamcpp

- ✓ De-risking fintech, blockchain, and AI platforms from strategy to production
- ✓ Architected RWA exchange, settlement, custody, and treasury systems
- ✓ Hands-on CTO and founder; led multiple teams of 6-10 with cross-team architectural oversight
- ✓ Supported filings for FMA/DORA, CertiK smart-contract audits, and ISO 27001 adoption
- ✓ Building self-hosted AI platforms end to end
- ✓ Published author in systems programming (*Extreme C*, Packt)

SUMMARY

Hands-on systems architect with 15+ years across startups, large corporations, and consulting, as engineer, Tech Lead, CTO, board member, and founder, pairing architectural leadership with building the critical components myself. I advise CTOs, chief architects, and tech leads on architecture, security, and regulatory compliance, most recently for tokenized-securities (RWA) infrastructure (on-chain exchange and auction engines with atomic settlement, a central securities depository, custody, and treasury), and build self-hosted AI platforms (LLMs, diffusion models, LoRA training). Core stack: C, C++, Golang, and Rust on Kubernetes and AWS/GCP.

WORK EXPERIENCE

DeKrypt B.V.

Founder, CEO & CTO, Principal Architect & Consultant
March 2024 - Present
Amsterdam, The Netherlands

- Consultancy: Delivering all part-time architecture engagements listed in this CV through the firm (freelance before its founding).
- SoulForge (own startup): AI persona platform: self-hosted LLM/VLM inference serving, RAG memory on a vector database, FLUX image generation, per-persona LoRA training, face-identity conditioning, elastic GPU nodes.
- FamBuzz & Bo2Bo2 (own startups): A private group social app and a mobile fitness game: Golang backends on Kubernetes via Terraform and CI/CD, Android/iOS clients.
- AI-Driven Development: Delivering with AI coding agents (Claude, Codex, Gemini), keeping accountability with engineers.
- FinTech & Distributed Ledger: Architecting RWA exchange, settlement, MPC custody, and DeFi treasury systems for Vero Labs, Assetera, and Metapolitan.
- Writing & Research: Authoring books and essays on software engineering, plus academic work for peer review.

Vero Labs Inc.

Senior Blockchain Architect (Part-time consultancy)
April 2024 - September 2026
New York, United States

↳ **Tokenise Ltd**

Senior Blockchain Architect (Part-time consultancy)
London, United Kingdom

↳ **VeroPulse GmbH**

Chief Software Architect & Tech Lead (Part-time)
Vienna, Austria - led a team of 6

- Agora Stock Exchange Engine: Architected the RWA exchange's on-chain (DeFi) orderbook and settlement engine, with atomic settlement finality between the orderbook and treasury contracts.
- Tokenise CSD: Designed the Central Securities Depository for on-chain issuance, MPC-wallet custody, and settlement of tokenized securities (RWA).
- Tokenise Broker: Architected the brokerage platform connecting investors to the Agora Engine for order routing and execution.
- Vero Android App: Directed the architecture of the primary and secondary market mobile client, integrated end-to-end with the Agora Engine.
- Platform & Infrastructure: Designed the off-chain Golang microservices, deployed on Kubernetes (AWS EKS) via Terraform and CI/CD for scalable, highly-available trading workloads.
- Regulatory Compliance: Contributed technical responses to filings for the FMA under DORA; the platform drew ESMA's interest, leading to a direct meeting.
- AI-Assisted Engineering: Introduced AI-assisted coding workflows (LLM copilots, automated code review) across the teams to accelerate delivery and improve code quality.
- Golang - Kubernetes (EKS) - AWS - Terraform - Solidity - EVM chains - PostgreSQL - Docker - gRPC

Assetera GmbH

Senior Blockchain Architect & Tech Lead (Part-time)
April 2024 - March 2026
Vienna, Austria

- Agora Auction Room: Architected the on-chain (DeFi) auction mechanics for RWA tokens, with atomic settlement finality enforced against the on-chain treasury.
- Tokenization as a Service: Architected the on-chain framework for tokenizing real-world assets (RWA): issuance, MPC-wallet custody, and transfer.
- Platform & Infrastructure: Designed the off-chain Golang services, deployed on Kubernetes (AWS EKS).
- Golang - Solidity - EVM chains - Kubernetes (EKS) - AWS

Metapolitan Ltd.

Chief Blockchain Architect (Part-time consultancy)
January 2024 - March 2026
London, United Kingdom

- Distributed Treasury System: Architected a multi-signature and MPC-wallet on-chain treasury platform for RWA and DeFi clients, enabling secure custody, automated settlement, and policy-driven fund controls across multiple blockchain networks.
- Solidity - Gnosis Safe (multi-sig) - MPC wallets - EVM chains - web3.js - Hardhat - Node.js

Digital First GmbH

Senior Blockchain Architect (Part-time consultancy)
December 2021 - March 2024
Vienna, Austria

- TheKiss NFT Tokenization Platform: Designed the NFT minting architecture and smart contract framework underpinning the platform's digital collectibles.
- ARTEQ Token: Architected the ERC-20 token economics and smart contracts for the art-backed asset; passed CertiK audit.
- QLINDO Token: Led smart contract architecture and security review for the token launch; passed CertiK audit.
- Hergé NFTs: Architected the ERC-721 minting and royalty framework for the licensed NFT collection.
- Solidity - EVM chains - ERC-20 - ERC-721 - Hardhat - OpenZeppelin

ActiveVideo B.V. *(April 2019 - February 2023)*

↳ **Commscope Inc.** *(March 2023 - March 2024)*

↳ **ARRIS B.V.** *(April 2024 - December 2025)*

↳ **Vistance Networks B.V.** *(January 2026 - Present)*

Senior Software Engineer
Hilversum, The Netherlands - one company, many reorganizations

- CloudTV (Video Stitching Engine): Low-latency cloud renderer/streamer delivering UI and audio/video to set-top boxes.
- Media Pipelines: Built audio/video pipelines on MPEG-TS and H.264 with ffmpeg, including CEA-608/708 closed captions carried in SEI.

• AppCloud: Containerized Android streaming platform with a Kubernetes (AWS EKS) backend, provisioned with Terraform and shipped through CI/CD.

• Multiview Engine: High-throughput cloud streamer broadcasting multiple channels at once to set-top boxes and TVs.

• Tech Council & Security Guild: Prioritized tech debt; drove threat modeling, secure SDLC, and ISO 27001 adoption across teams.

• C++17 (`g++`, `clang`) - Golang - Java and Kotlin - Python - AWS - Kubernetes (EKS) - S3 - EFS - EBS - Ceph - Envoy

TomTom B.V.

Senior Software Engineer

September 2017 - March 2019

Amsterdam, The Netherlands

• Expert Software Consultant from Modis B.V.

• NavKit Free Text Search Engine, NavKit Worker REST Services, POI Info Services

• C++11 (`g++`, `clang`) - Java SE - JUnit - Spring MVC - JAXB - Spring REST - `gdb` - Valgrind - Clang ASAN & TSAN - STL & Boost - Google Test - Protobuf - Thrift - Python - CI/CD (Jenkins) - Docker - JIRA - Fisheye - Confluence - Bitbucket Stash - AWS/Azure based cloud

Jeppesen Systems (a Boeing subsidiary)

Senior Software Engineer

February 2017 - August 2017

Gothenburg, Sweden

• Manpower Application

• C++11 (`g++`) - Python - C#.NET - `gdb` - Valgrind (callgrind, massif) - PDB - STL & Boost - Google Test - XMLRPC - Jenkins - JIRA - Fisheye - Confluence

Adro Co.

Senior Software Architect, Principal C++ Engineer

June 2016 - January 2017

Tehran, Iran

• Low-latency, high-throughput Realtime Bidding (RTB) Engine

• C++ (`g++`) - Python - `gdb` - Valgrind - STL & Boost - libuv - Google Test - Protobuf - Jenkins - Travis CI - Docker - JIRA - Fisheye

Naeem Co.

Chief Technical Officer, Chief Software Architect

April 2015 - October 2016

Tehran, Iran

• Projects: ePassport Issuance System - Face Recognition System, Biometric Identification System

• Systems & Security: Performance-critical C/C++ services; threat modeling and secure SDLC for PKI systems

• Cryptography: OpenSSL - PKCS (1, 7, 10, 11, 12) - ICAO MRTD standards - HSM - Key Management - PKCS11 Tokens

• Stack: C (`gcc`) - C++ (`g++`) - Java SE - Java EE - Spring MVC - Wildfly - Tomcat - Python - `gdb` - Valgrind - STL & Boost - wxWidgets - Qt - Google Test - Jenkins - Docker - vSphere - Proxmox - Gluster - Cassandra - Redmine - GitLab

Sharif CERT Center

Software Engineer, CERT Team Member

December 2011 - June 2013

Tehran, Iran

• Member of the Computer Emergency Response Team (CERT): incident response, threat analysis, and building the tooling used to detect and contain network threats.

• Botnet Detection System - Sinkhole - DDNS as a FreeBSD module

• C++ (`g++`) - Java SE - Perl - `gdb` - JConsole - Valgrind - STL & Boost - Swing - Redmine

TECHNICAL SKILLS

• **Languages & Architecture:** C++ (14/17/20), C, Golang, Rust, Python, Java, JS - Distributed systems (CAP, replication, leader election, CQRS, event sourcing, message queues, streaming, async/event loops) - Low-latency, high-throughput design

• **Cloud & DevOps:** Kubernetes (`kubectl`, Helm, Kustomize, PSA) - AWS & GCP (EKS/GKE, EC2/GCE, S3, VPC, IAM) - Terraform, CI/CD - Docker - PostgreSQL, MySQL, MongoDB, Redis

• **Blockchain:** Solidity (EVM), Solana, web3.js/ethers.js, Hardhat - Distributed ledgers, smart contract architecture, RWA, DeFi, MPC wallets

• **Security & Cryptography:** OpenSSL, PKI, symmetric/asymmetric algos, hashing/MAC - Protocols & standards (gRPC, Protobuf, REST, PKCS, RFCs) - Networking (TCP/IP, Wireshark, `tcpdump`) - Threat modeling, secure SDLC, incident response - Compliance (DORA, ISO 27001) - CISSP (in progress)

• **AI & LLMs:** Self-hosted LLM/VLM & diffusion pipelines (Qwen, FLUX, Diffusers, Ollama) - LoRA training, identity conditioning (PuLID/IP-Adapter, face embeddings), inpainting, prompt compilation, RAG and agent memory (vector databases), elastic GPU inference serving - Classical ML & computer vision (TensorFlow, OpenCV, CUDA) - Agentic coding & review with Claude, Codex, and Gemini

• **Media & Streaming:** Audio/video pipelines - MPEG-TS, H.264, ffmpeg - Closed captions (CEA-608/708 in SEI)

• **Embedded & Kernel Development:** Linux kernel modules & system calls - FreeBSD kernel modules - Low-level C/C++ (memory, concurrency, IPC)

PUBLICATIONS & OPEN SOURCE PROJECTS

• **Extreme C:** Book published October 2019 by Packt Publication.

• **Linux Two-Step Authentication with Rust:** liveProject for Manning Publications, with its accompanying ESS backend and client projects in Rust.

• **Other Rust projects:** Domain Catcher Engine and the Outlook .msg parser.

LANGUAGES

Persian (**Native**), English (**Fluent**), German (**A1**), Dutch (**B1**), French (**A1**)

EDUCATION

• **Bachelor of Science in Physics (Astronomy and Planetary Science)**, The Open University, started 2017 (3 years completed, on leave)

• **Bachelor of Software Engineering**, University of Isfahan (Isfahan, Iran), 2004 - 2008

PERSONAL INTERESTS

• **Physics and Astronomy**: Cosmology, Quantum and Particle Physics, Relativity, Photonics

• **Mathematics**: Algebra, Probability Theory, Information Theory

• **Cryptography**: Information and Network Security, Quantum Cryptography and Key Distribution

• **Philosophy**: Western Philosophy, Sociology, Economics, Politics

• **Telecommunications**: Modulation, Multiplexing, Optical Fibers, Coding Theory